
Designing of Database

Entity-Relationship Model

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How to design the database?

There are two approaches

- E-R Modeling: Identifying entity and relations
- Normalization: Refinement of database designing

Database Design Techniques

- Top down Approach
 - ER Modeling
 - Bottom Up Approach
 - Normalization
-

Entity-Relation Model

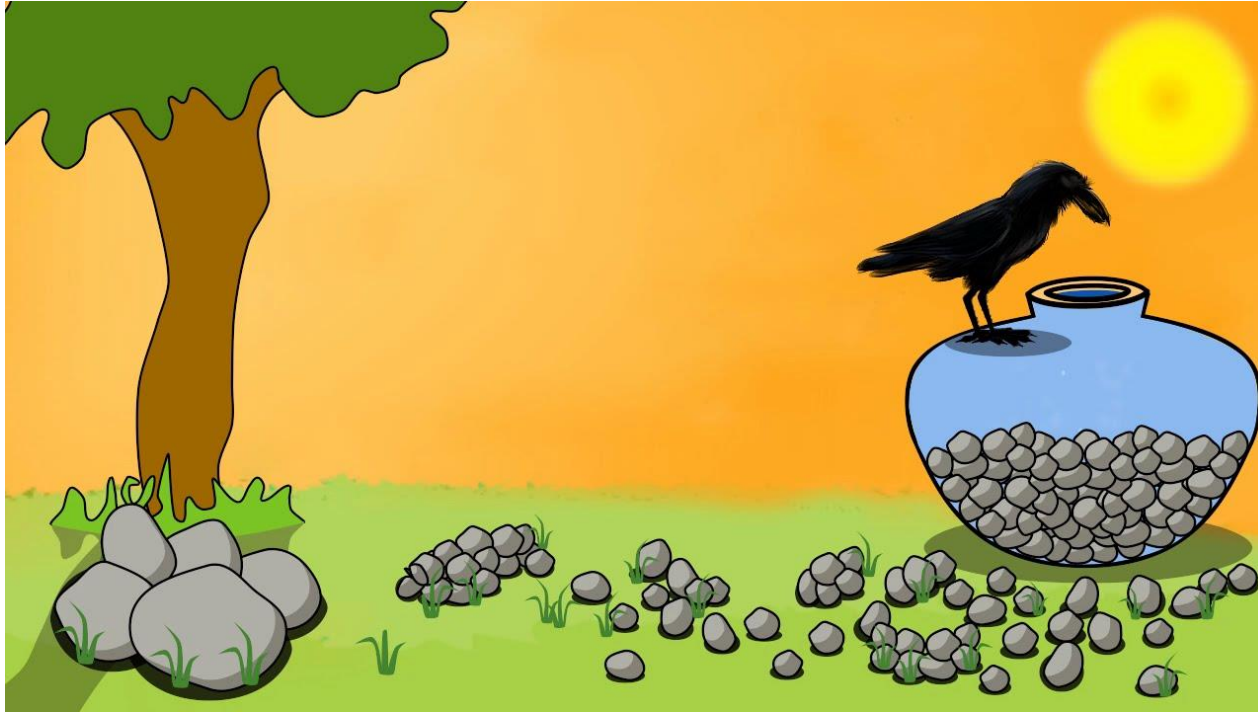
E-R Model

- The Entity-Relationship (ER) model was originally proposed by **Peter Chen** in 1976
- ER model is used to visually represent data objects.
- The ER model is a conceptual data model that views the real world as entities and relationships.

Need of ER Model

- ER Model is used to convert software requirements written in textual form into visual representation.
- Because
 - One picture is equivalent to thousand words.

What comes to your mind when you see this..



Case Study: University Management System

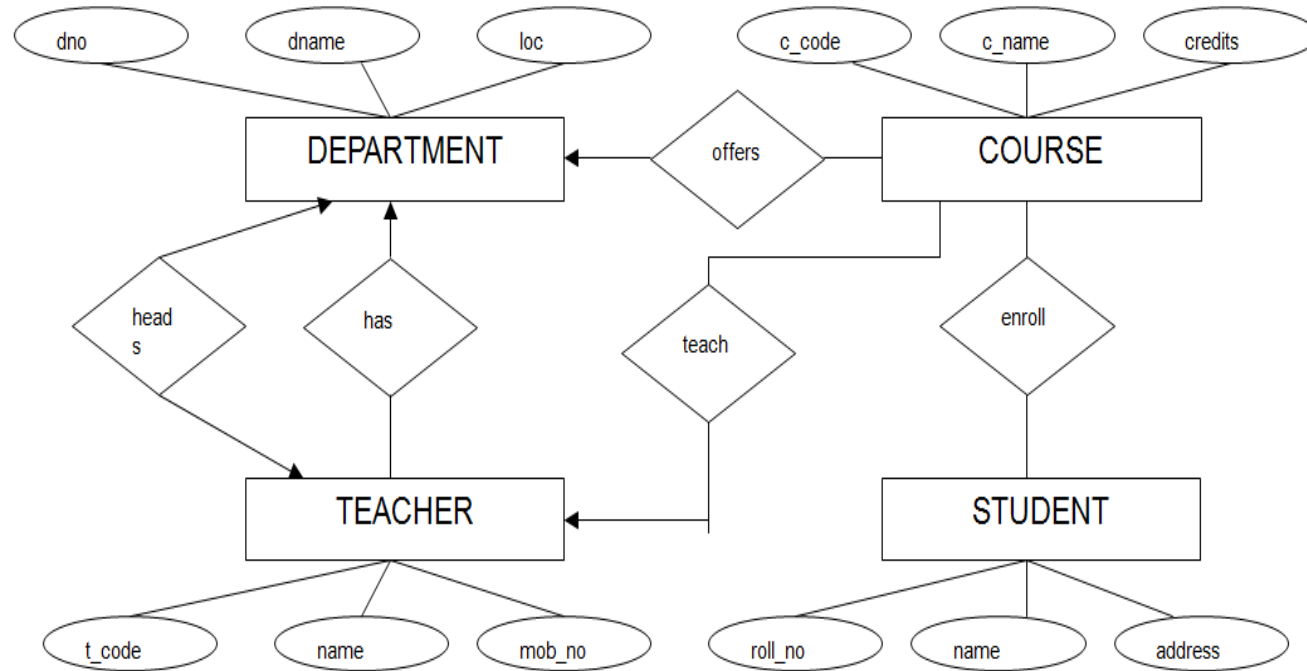
- Consider, a university contains many departments. Each department can offer any number of courses. Many teachers can work in a department. A teacher can work only in one department. For each department there is a Head. A teacher can be head of only one department. Each teacher can take any number of courses. A course can be taken by only one instructor. A student can enroll for any number of courses. Each course can have any number of students.

Issues in Capturing requirements in plain text form

- Boring
 - Difficult to capture all the details
 - Error prone
 - Difficult to understand
 - Difficult to design database from plain text
-

Solution

Use pictures instead of textual explanation



ER Modeling

- A graphical technique for understanding and organizing the data.
- ER model is used to visually represent data objects
- It is independent of the actual database implementation.

Basic Constructs of E-R Modeling

- An **entity** is an object whose information is stored in the databases. Every entity is unique and distinguishable from other entities/objects.
 - Example: specific person, company, event, plant
- An **entity set** is a set of entities of the same type that share the same properties.
 - Example: set of all persons, companies, trees, holidays
- Entities have ***attributes***
 - Example: people have *names* and *addresses*

Basic Constructs of E-R Modeling

- A *database* can be modeled as:
 - a collection of entities,
 - relationship among entities.

Correlation between different models

Class Diagram	ER Model/ER Diagram	Relational Model
Class	Entity Set	Table
Object	Entity	Row
Attribute	Attribute	Column

Relationship

- A Relationship represents an association between two or more entities.

Examples

- Employees belong to department
- Customer has taken loans

Entity Sets *customer* and *loan*

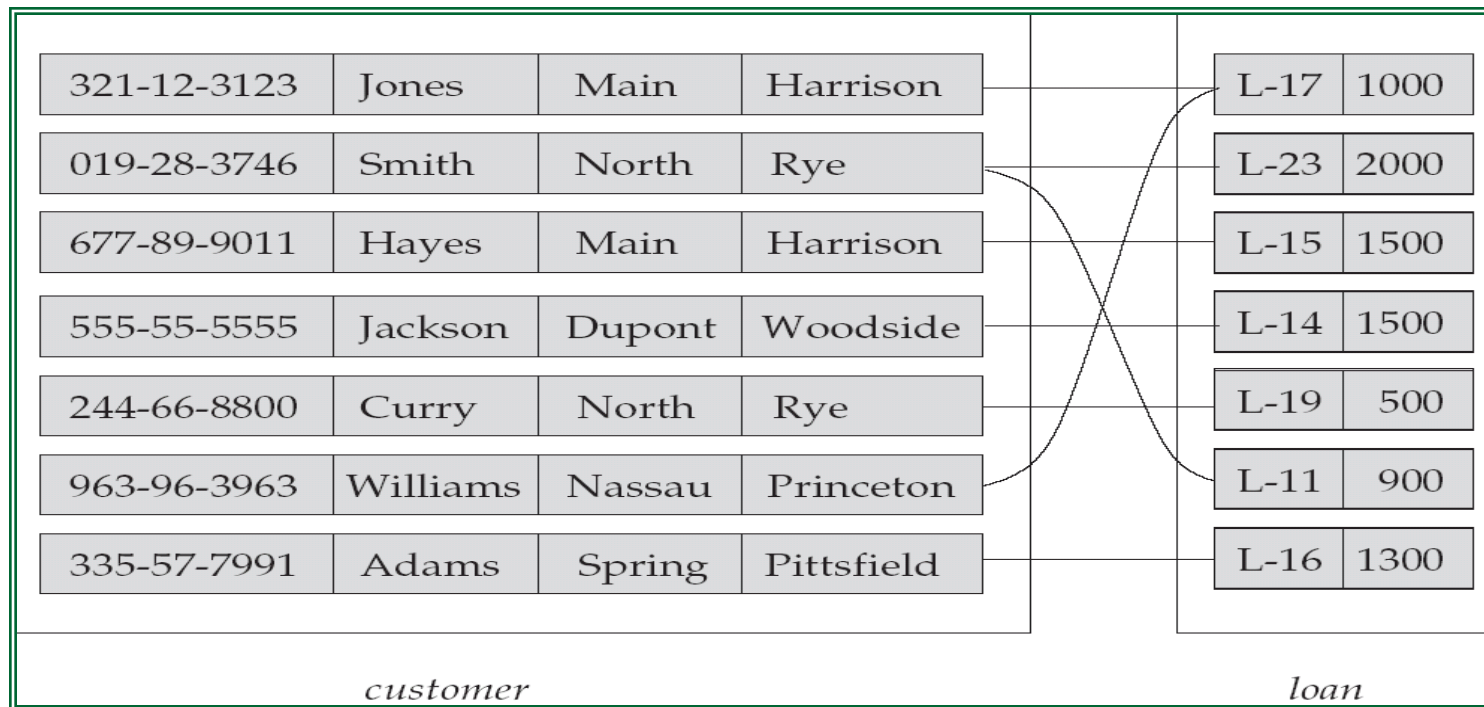
321-12-3123	Jones	Main	Harrison
019-28-3746	Smith	North	Rye
677-89-9011	Hayes	Main	Harrison
555-55-5555	Jackson	Dupont	Woodside
244-66-8800	Curry	North	Rye
963-96-3963	Williams	Nassau	Princeton
335-57-7991	Adams	Spring	Pittsfield

customer

L-17	1000
L-23	2000
L-15	1500
L-14	1500
L-19	500
L-11	900
L-16	1300

loan

Relationship Set *borrower*



Attributes

Attributes describe the properties of the entity of which they are associated.

- Simple
- Composite
- Single-values
- Multi-values
- Stored
- Derived

Simple Vs Composite Attribute

- Simple Attribute
 - ❑ Cannot be divided into simpler components
 - ❑ Examples: Age of employee, Roll number of students
- Composite Attribute
 - ❑ Can be divided into simpler components
 - ❑ Examples: Name of employee can be First Name and Middle Name; Address of Student can be House Number, Locality

Stored Vs Derived Attribute

■ Stored Attribute

- ❑ Attribute that need to be stored permanently
- ❑ Example: Name of student, Roll Number of student

■ Derived Attribute

- ❑ Attribute that can be calculate based on other attributes.
- ❑ Example: Age can be calculated based on Date of Birth

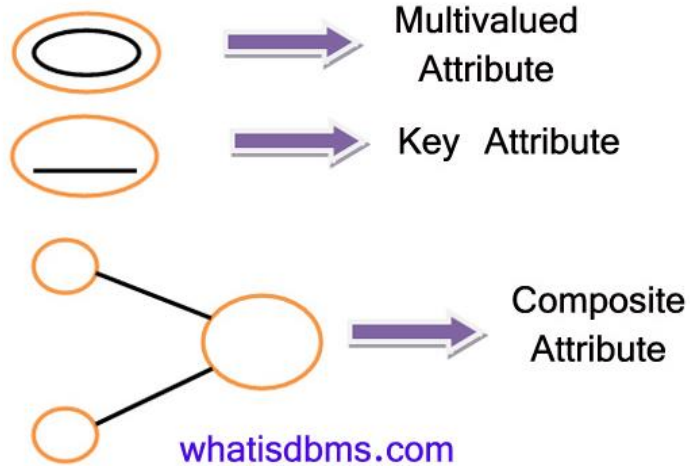
Symbols used to represent these constructs

  Entity Set

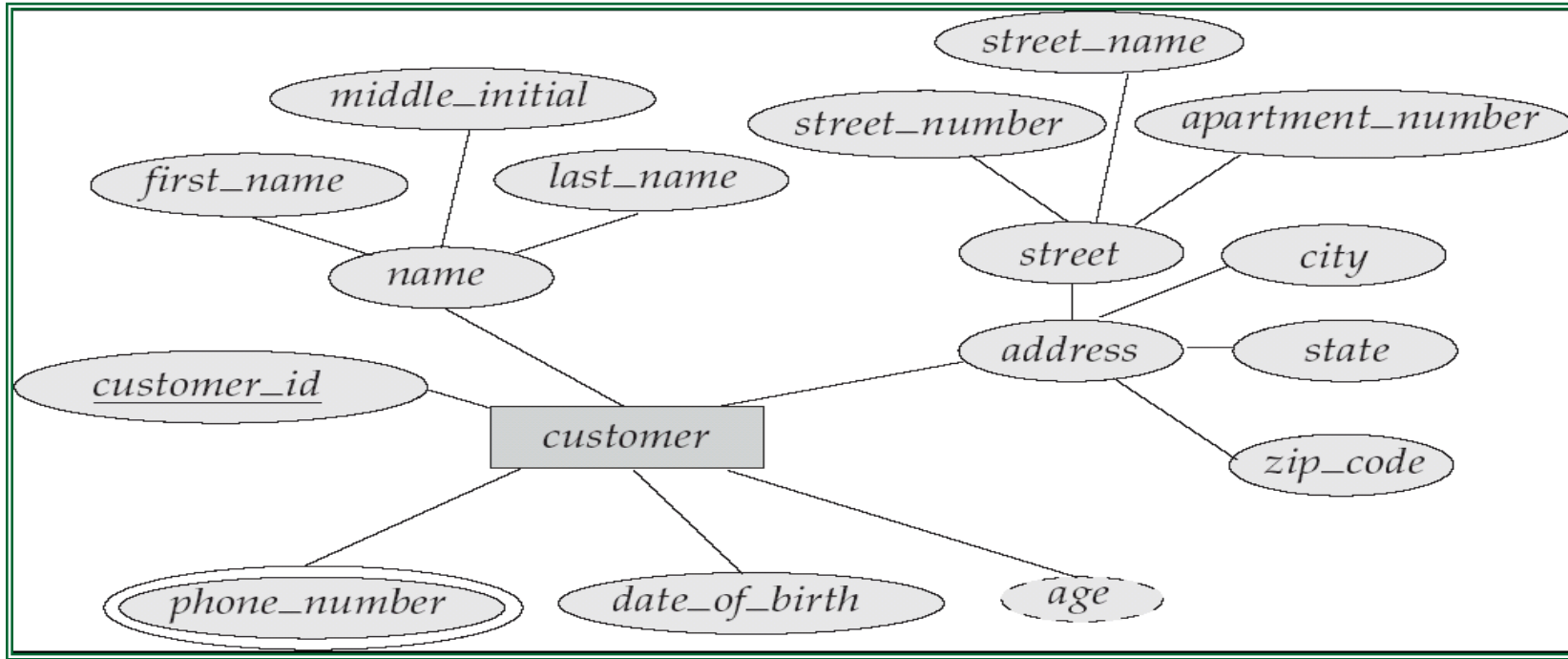
  Relationship

  Attribute

Symbols used to represent these constructs



Example



Degree of a Relationship

- The degree of a relationship is the number of entities associated with the relationship.
- The n -ary relationship is the general form for degree n . Special cases are the binary, and ternary, where the degree is 2, and 3, respectively.

Connectivity of a Relationship

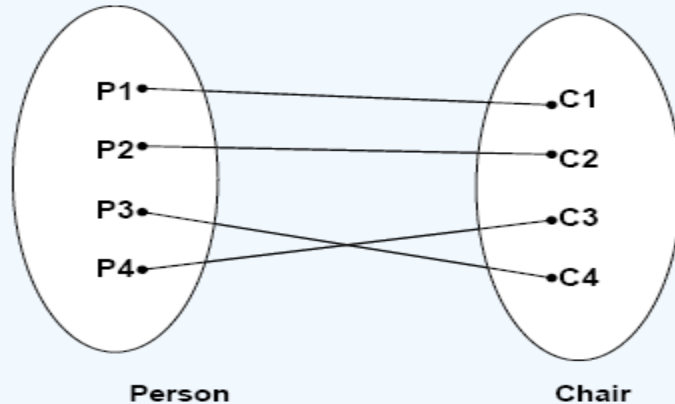
The connectivity of a relationship describes the mapping of associated entity instances in the relationship. The values of connectivity are "one" or "many".

The basic types of connectivity for relations are:

- One to One (1:1)
- One to Many (1:M)
- Many to One (M:1)
- Many to Many (M:M)

Connectivity of a Relationship

One - To - One

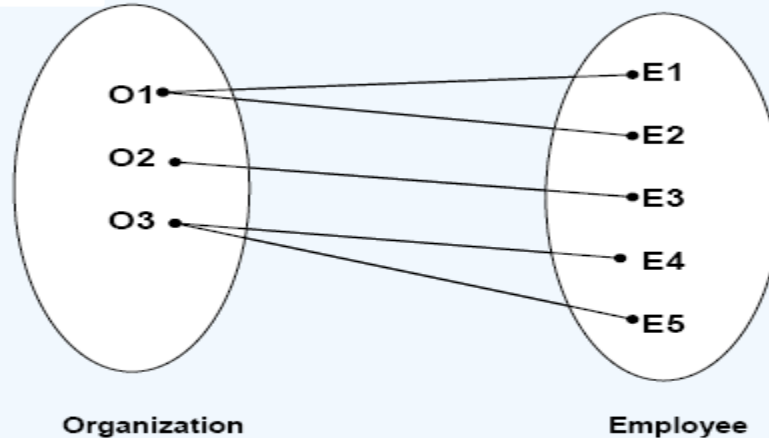


One instance of entity type Person
is related
to one instance of the entity type Chair.



Connectivity of a Relationship

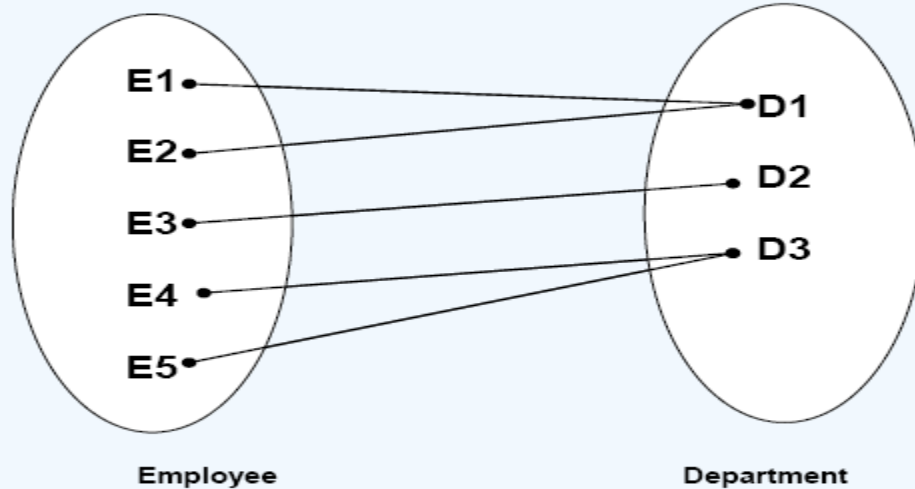
One -to- Many



One instance of entity type Organization
is related
to multiple instances of entity type Employee

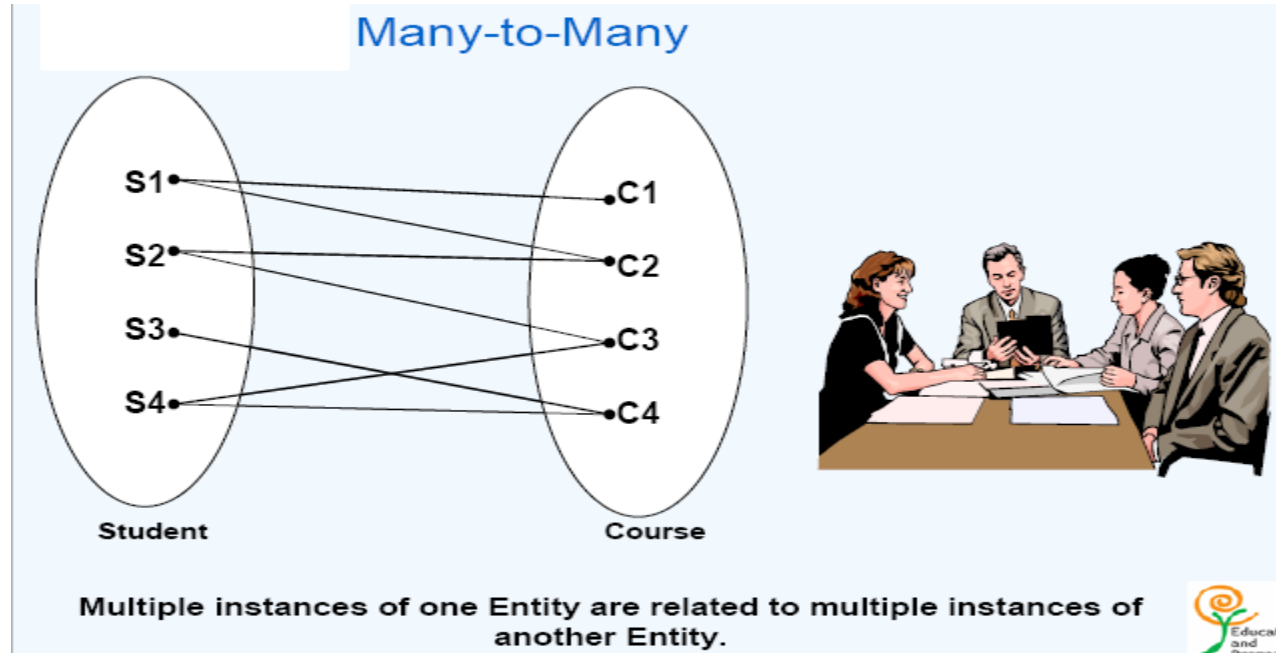
Connectivity of a Relationship

Many-to-One



Reverse of the One to Many relationship.

Connectivity of a Relationship



Find out the type of Connectivity

- Student and Teacher Entity Sets
- Answer: Many to Many
- Student and Department Entity Sets
- Answer: Many to One
- College and Society Entity Sets
- Answer: One to Many
- Team and Captain Entity Sets
- Answer: One to One

Direction

- The direction of a relationship indicates the originating entity of a relationship.
- The entity from which a relationship originates is the parent entity; the entity where the relationship terminates is the child entity.
- The type of the relation is determined by the direction of line connecting relationship component and the entity.

Direction

- To distinguish different types of relation, we draw either a directed line or an undirected line between the relationship set and the entity set.
- Directed line is used to indicate one occurrence and undirected line is used to indicate many occurrences in a relation.

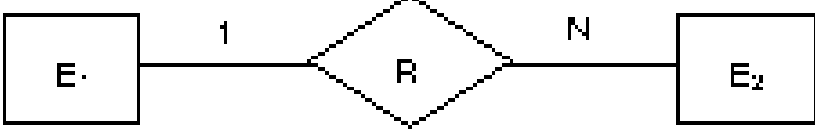
Directed line



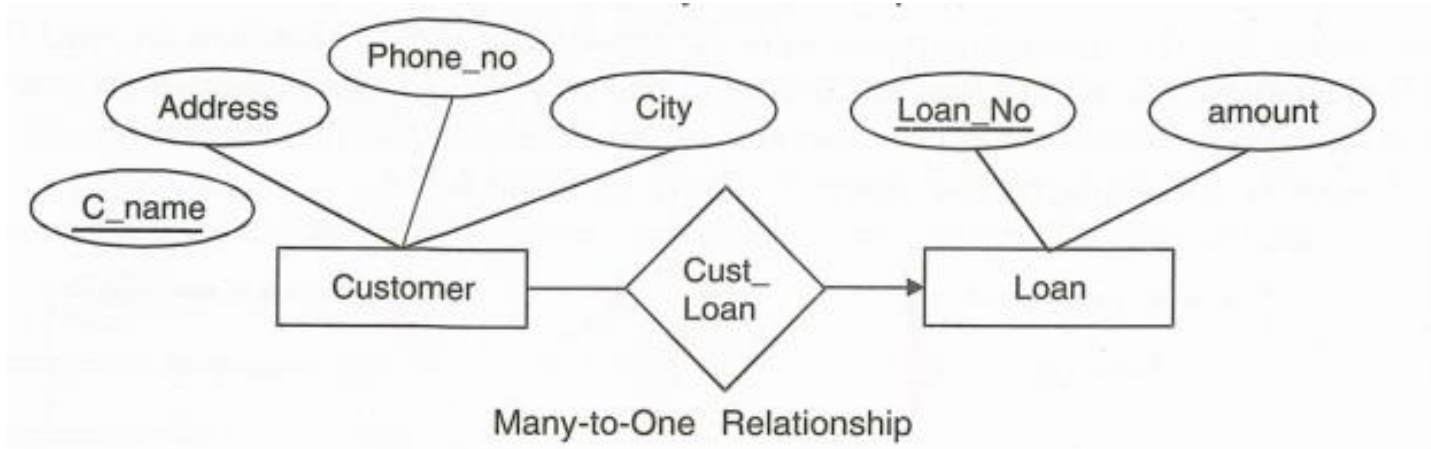
Undirected line



Symbol of Relationship

	Relationship Type
	Cardinality Relation 1:N for E1:E2 in R

Customer-Loan Relationship



Designing an ER Model

Supplier Part Case Study

- Design ER model for Supplier part database, where one supplier can supply multiple parts and for a give part there are multiple suppliers. Whenever supplier supplies some part we have to maintain the qty in which part is supplied. Every Supplier has unqiue Sno, and other field of interest are Name and City. Sameway, every part has unique Pno and other field of interest are Name and Color.
-

Supplier Part Database

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- Every Supplier has unqiue Sno, and other field of interest are Name and City.
- Sameway, every part has unique Pno and other field of interest are Name and Color.

E-R Model to capture relationship between Emp and Manager

Case Study: University Management System

- Consider, a university contains many departments. Each department can offer any number of courses. Many teachers can work in a department. A teacher can work only in one department. For each department there is a Head. A teacher can be head of only one department. Each teacher can take any number of courses. A course can be taken by only one instructor. A student can enroll for any number of courses. Each course can have any number of students.

Steps to design E-R diagram

- **Step 1 to Identify the Entities**
- **Step 2 to find relationships among these entities**
- **Step 3 to identify the key attributes**
- **Step 4 to identify other relevant attributes**
- **Step 5 to draw the complete e-r diagram**

First Step: To Identify the Entities

- In order to identify the entities collect all the noun in the requirement sheet which has some properties and are important for the system.

First Step: To Identify the Entities

We can identify the following nouns:

- University, Department, Course, Teacher, Student.
- Here, the database is of only one university. If an entity has a single instance then that entity is ignored. Thus, the final entities are:
- DEPARTMENT
- COURSE
- TEACHER
- STUDENT

Second Step:

To find relationships among these entities

- Relationship are indicated by verbs or roles in the requirement sheet.
- So, we have to read each line of requirement sheet one by one and identify the verb between the nouns or entity sets.

Second Step:

To find relationships among these entities

- Consider, a university contains many departments.
- Each department can offer any number of courses.
- Many teachers can work in a department. A teacher can work only in one department.
- For each department there is a role of Head which is performed by the Teacher. A teacher can act as head of only one department.

Second Step:

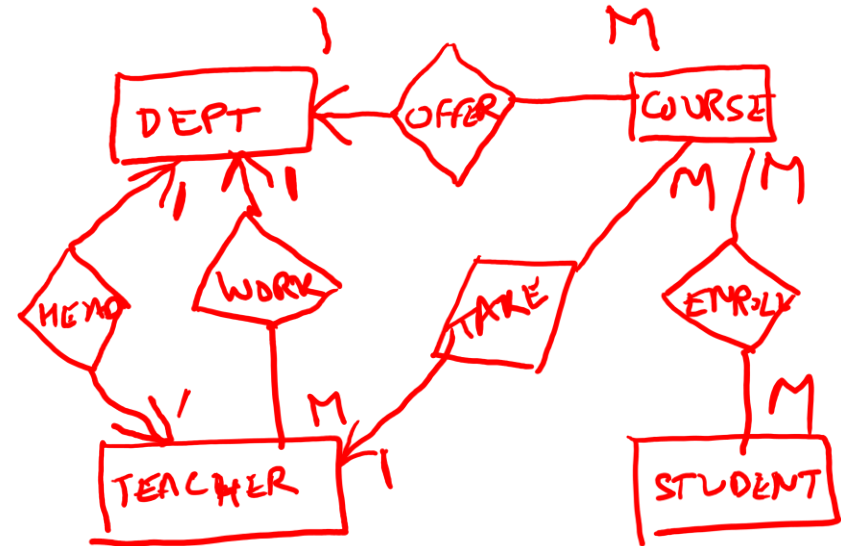
To find relationships among these entities

- Each teacher can take any number of courses. A course can be taken by only one instructor.
- A student can enroll for any number of courses. Each course can have any number of students.

Second Step:

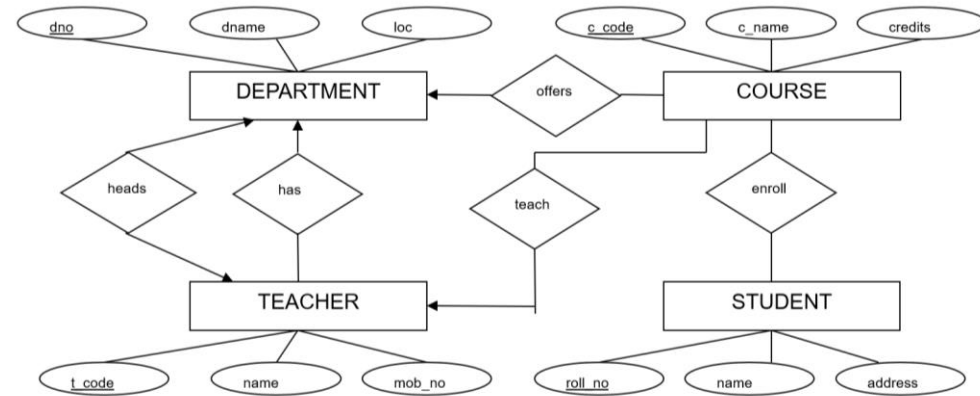
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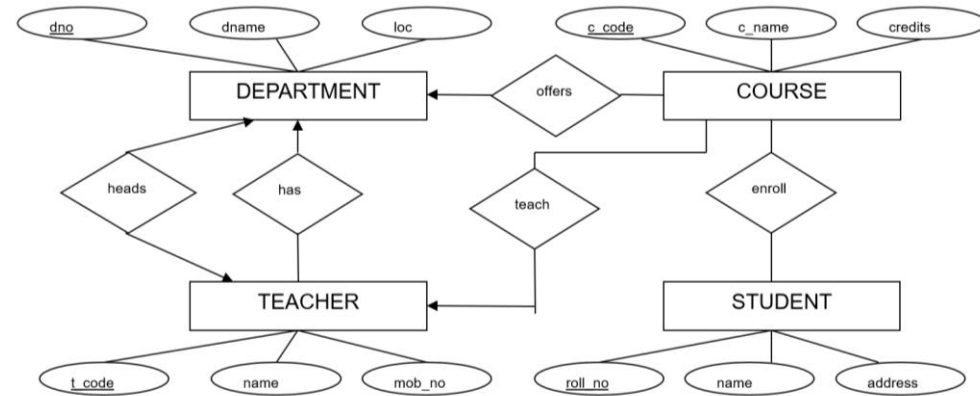
Step 3: To identify the key attributes

- Dno (Department number) is the key attribute for the Entity DEPARTMENT.
- C_code (Course number) is the key attribute for COURSE Entity.
- Roll_no (Roll number) is the key attribute for STUDENT Entity.
- T_code (Teacher code) is the key attribute for TEACHER Entity.

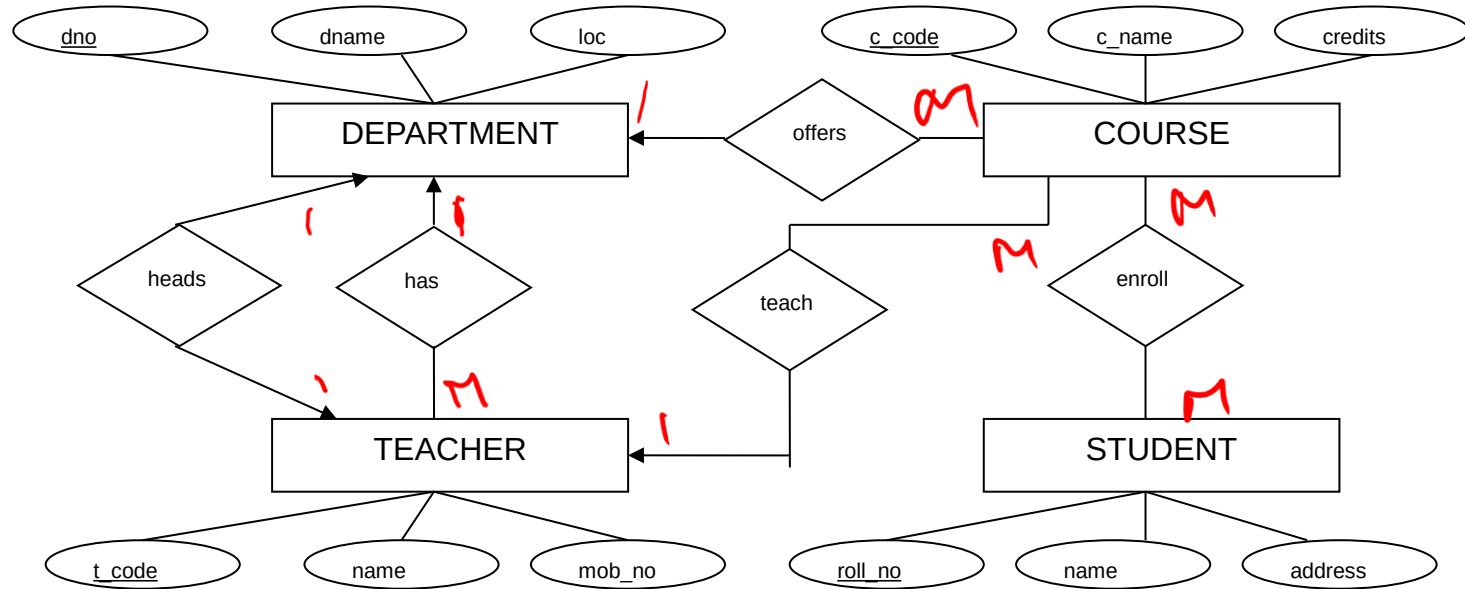


Step 4: To identify other relevant attributes

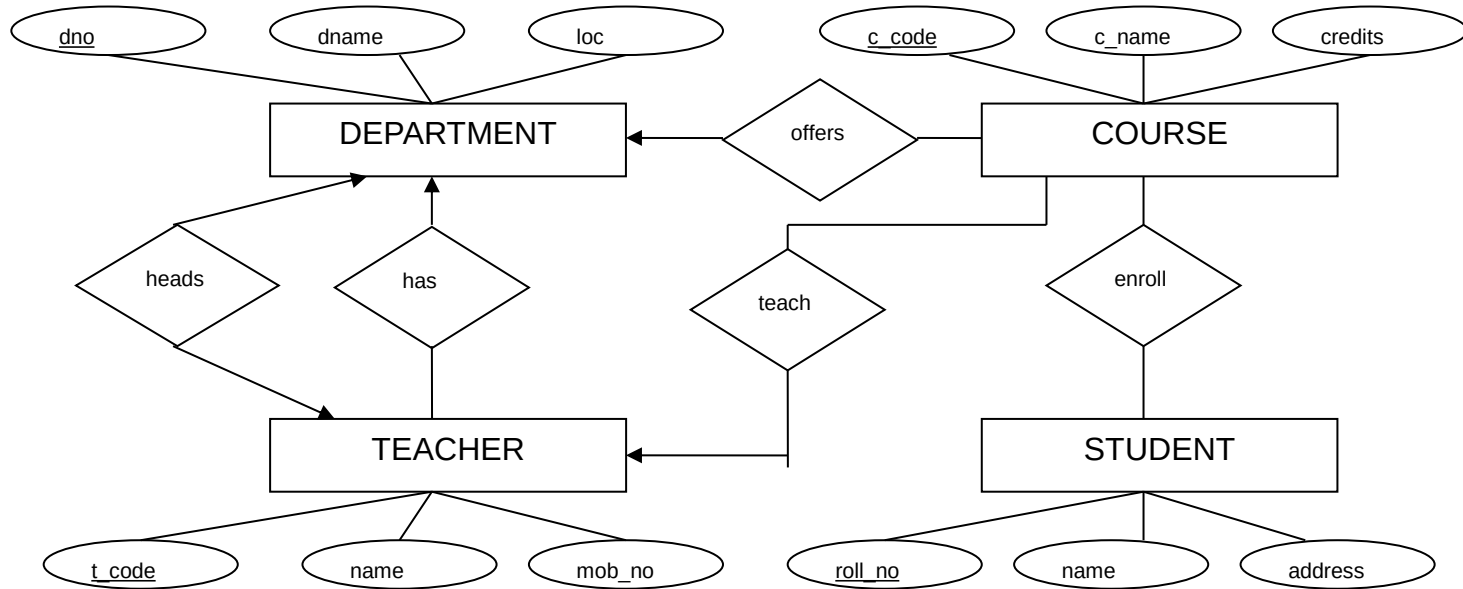
- DEPARTMENT entity will have other relevant attributes as dname, loc.
- For COURSE entity, c_name, credits.
- For TEACHER entity, name, mob_no
- For STUDENT entity, name, address



Step 5: To draw the complete ER diagram



Step 5: To draw the complete ER diagram



Class Activity

- Represent each of the following requirements with an ER diagram:
A regional council requires the design of a database system that can provide information on all schools in the region. The requirements collection and analysis phase of the database design process has provided the following data requirements for the schools database system.

(a) Every school has many students and many teachers. Each student is assigned to one school and each teacher works for one school only.

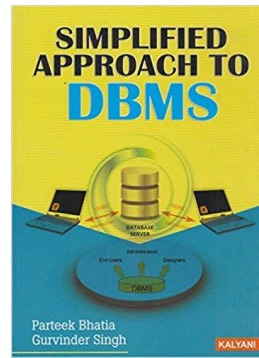
(b) Each teacher teaches more than one subject but a subject may be taught by more than one teacher. The database should store the number of hours a teacher spent teaching a subject. Data held on each teacher includes his/her national Insurance Number (NIN), name (first and last), sex, and qualifications. The data held on each subject includes subject title and type.

(c) Each student can study more than one subject and a subject may be studied by more than one student. Data held on each student includes the student's code, name (first and last), sex, and date of birth.

-
- (d) Each school is managed by one of its teachers. The database should keep track of the date he/she started managing the school. Data stored on each school includes the school's code, name, address (town, street, and post code) and phone
-

Conversion of E-R Diagram to Tables

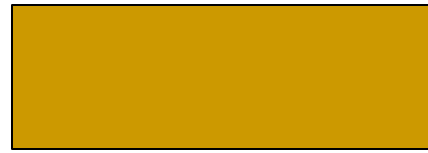
Handling of Entity Sets



Handling of Entity Sets

- For each entity set create a unique table.
- As entity set refers to class in OOP and table in relational model.
- Assign the name of table as name of the corresponding entity set.

Conversion of Entity Set to Relational Table



Entity Set



Relational Table

Conversion of Entity Set to Relational Table



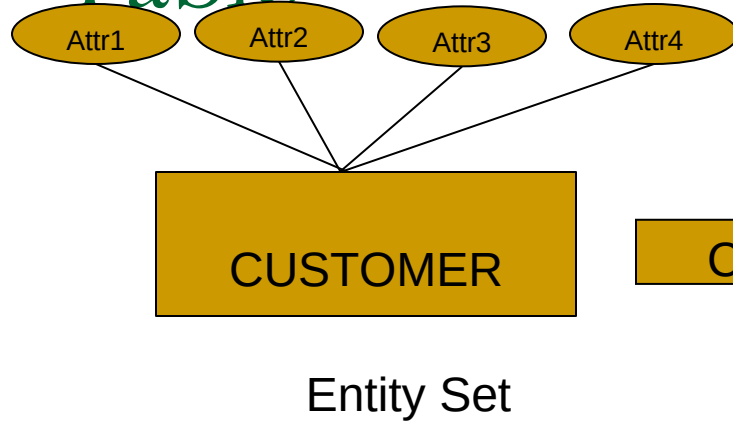
Entity Set



CUSTOMER

Relational Table

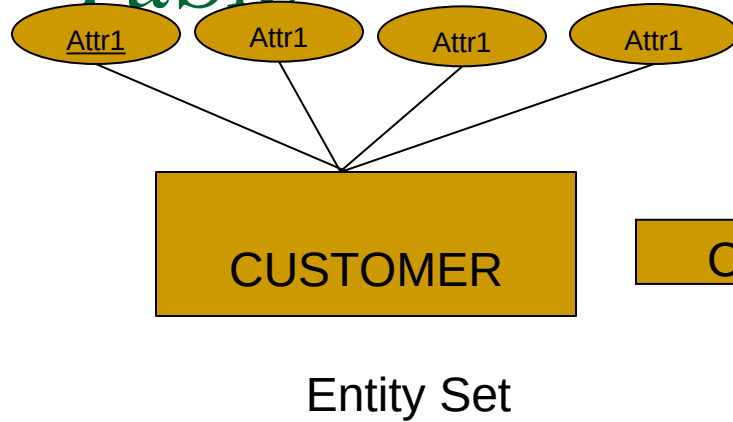
Conversion of Entity Set to Relational Table



CUSTOMER			
Attr1	Attr2	Attr3	Attr4

Relational Table

Conversion of Entity Set to Relational Table



Primary Key

CUSTOMER

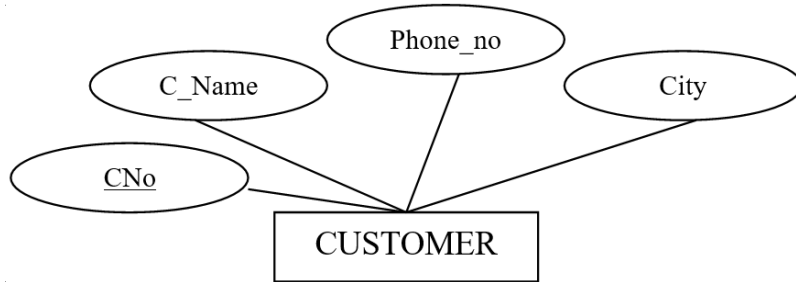
Attr1	Attr2	Attr3	Attr4

Relational Table

Handling of Attributes

- For every simple attribute create a column of the table.
- The key attribute will become primary key of the table.
- However, composite attribute, multi-value attribute and derived attribute need special treatment.

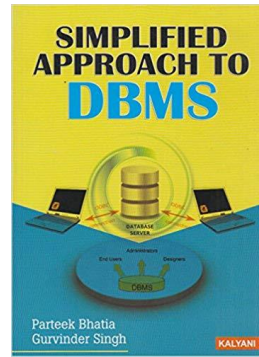
Example:



CNo	C_Name	Phone_No	City
100	Rahat Bhatia	2444566	Patiala
101	Ruhani Sharma	4547235	Jalandhar
102	Raj Singh	3445432	Amritsar

Conversion of E-R Diagram to Tables

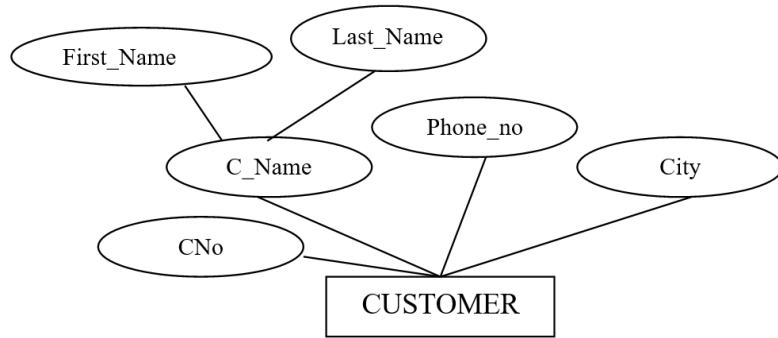
Handling of Composite Attributes



Handling of Composite Attribute

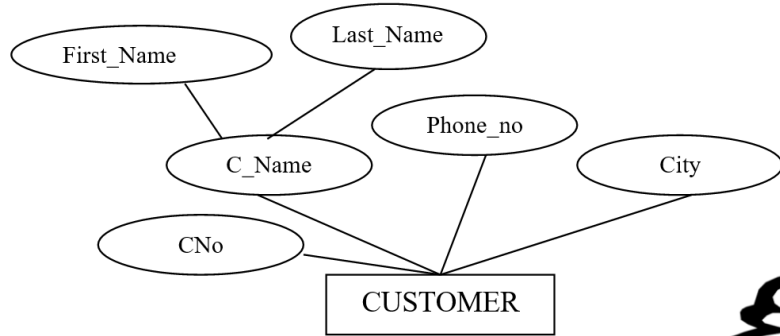
- Composite attributes are those which can split into components.
- For example: Name as First_Name and Last_Name
- In order to convert an entity set to table the composite attributes are flattened out by creating a separate attribute for each component of composite attribute.
- Create the columns for branches and ignore the root.

Handling of Composite Attribute



CNo	First_Name	Last_Name	Phone_No	City
100	Rahat	Bhatia	2444566	Patiala
101	Ruhani	Sharma	4547235	Jalandhar
102	Raj	Singh	3445432	Amritsar

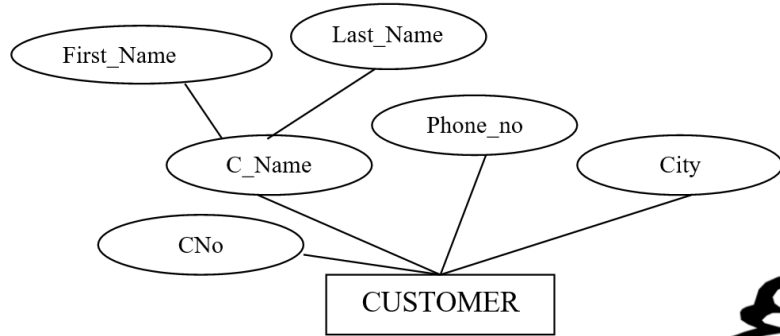
Handling of Composite Attribute



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Handling of Composite Attribute

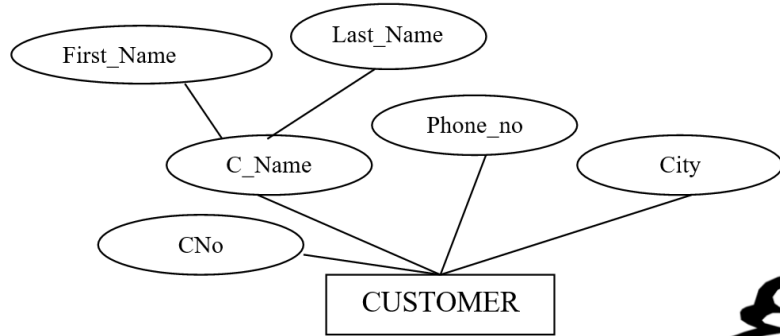


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100	Rahat	Bhatia	2444566	Patiala
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Solution is
Use of Concatenate
operator in SELECT
statement



Handling of Composite Attribute



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100	Rahat	Bhatia	2444566	Patiala
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The Query will be



```
SELECT CNO,  
FIRST_NAME|| ' '||LAST_NAME AS FULL_NAME  
FROM CUSTOMER;
```

Exercise

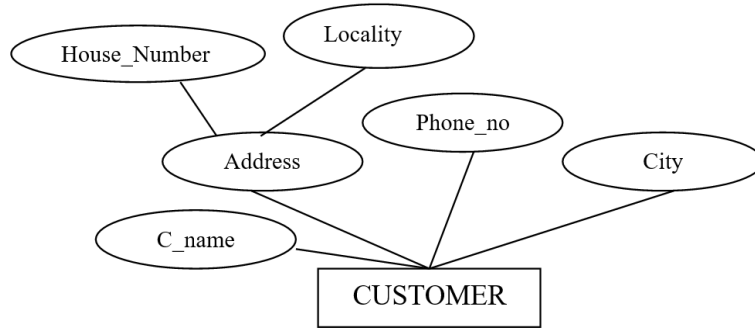
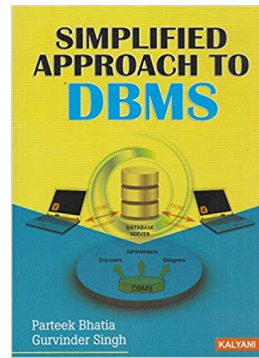


Table Customer

- Columns
- C_name, House_Number, Locality, Phone_no, City

Conversion of E-R Diagram to Tables

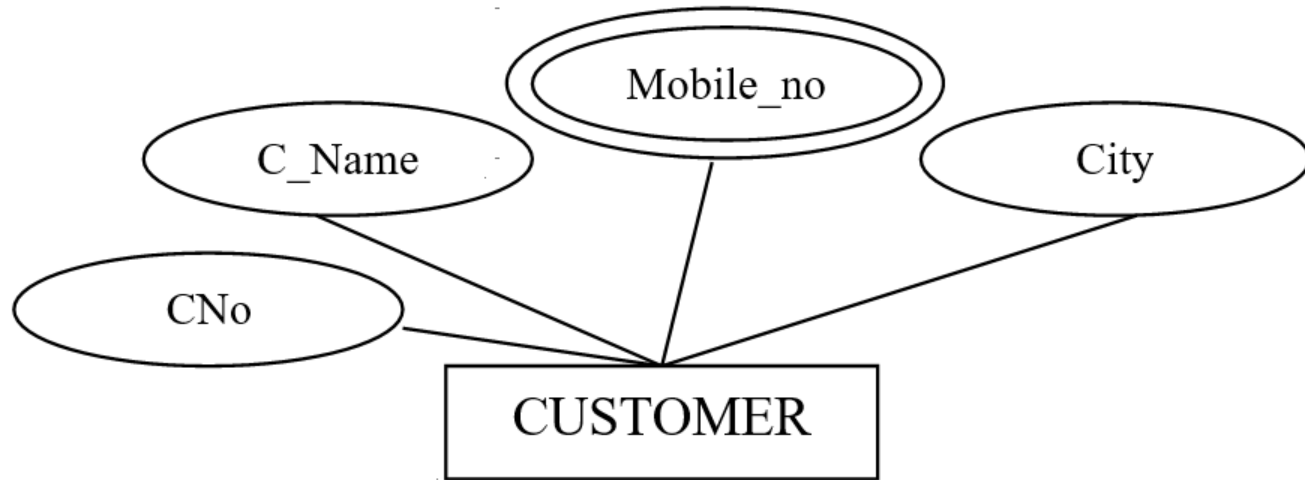
Handling of Multi-Value Attributes



Multi-Value Attributes

- Multi-value attributes are those which can take many values.
- For example: Mobile_Number, Hobbies, Skills etc.

Example



Possible Solution

Create a column for each possible value of attribute for a record.

CNo	C_Name	City	Mobile_No1	Mobile_No2	Mobile_No3
100	Rahat Bhatia	Patiala	1876115046	8739593711	8739593715
101	Ruhani Sharma	Jalandhar	3475784928		
102	Rishan	New Delhi	7457483929	8734648483	

Possible Solution: Create a column for each possible value of attribute for a record

Cno	C_Name	City	Mobile_No1	Mobile_No2	Mobile_No3
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102	Rishan	New Delhi	7457483929	8734648483	



```
SELECT C_NAME FROM CUSTOMER WHERE  
MOBILE_NO1=8739593711  
OR  
MOBILE_NO2=8739593711  
OR  
MOBILE_NO3=8739593711;
```

Possible Solution: Create a column for each possible value of attribute for a record

Cno	C_Name	City	Mobile_No1	Mobile_No2	Mobile_No3
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First we have to find that in which column 8739593711 is stored and then we have to update it. Will be very tough and tricky. Simplicity of RDBMS is lost...

Second Solution

Store the multiple values of the column by comma separation.

Cno	C_Name	City	Mobile_Nos
100	Rahat Bhatia	Patiala	1876115046, 8739593711, 8739593715
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Integrity of data is lost.
Still Retrieve and Updation and
Deletion operations are tricky
Will be very tough and tricky.
Simplicity of RDBMS is lost...

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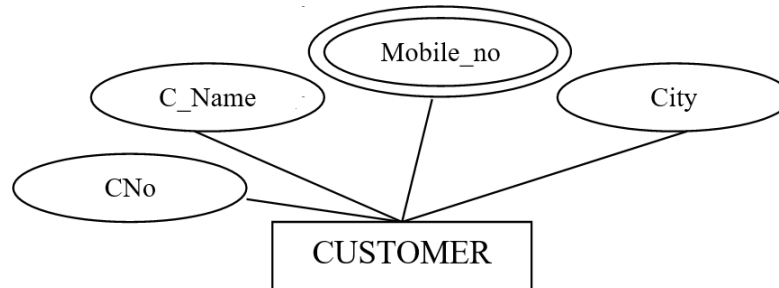
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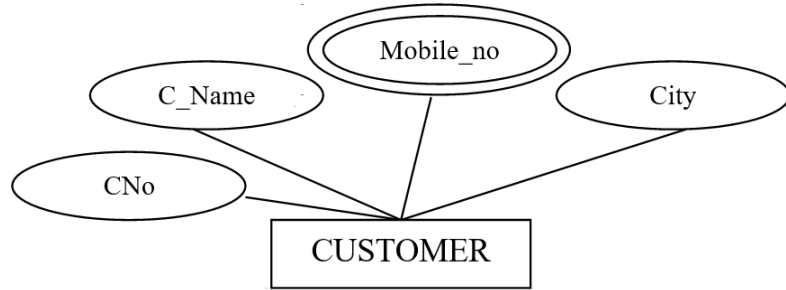
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Solution for Handling of Multi-value Attribute

- A multivalued attribute M of an entity E is represented by a separate table EM.
- A table EM has attributes corresponding to the primary key of E and an attribute corresponding to multivalued attribute M.



Handling of Multi-value Attribute

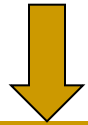
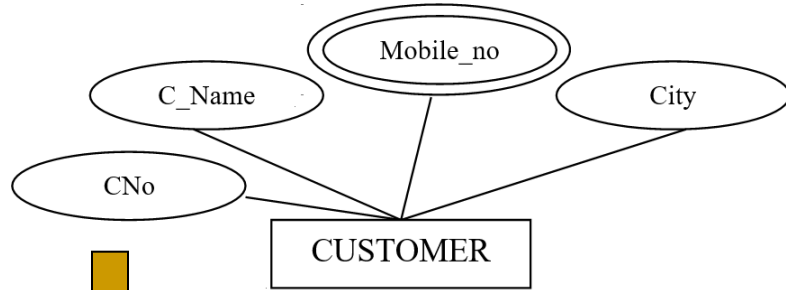


- Solution
- Two Tables

Cno	C_Name	City
100	Rahat Bhatia	Patiala
101	Ruhani Sharma	Jalandhar
102	Rishan	New Delhi

Cno	Mobile_No
100	1876115046
100	8739593711
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Handling of Multi-value Attribute



Primary Key

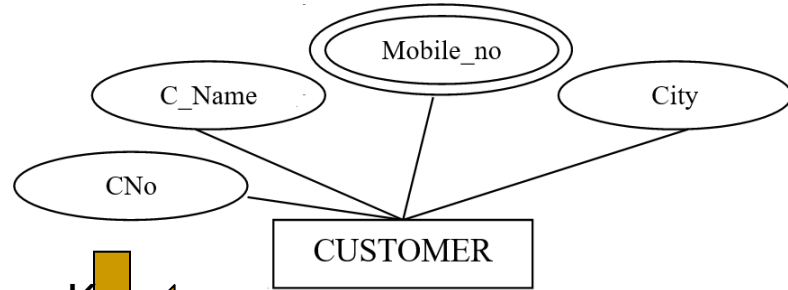
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- Solution
- Two Tables

Primary Key

Cno	Mobile_No
100	1876115046
100	8739593711
100	8739593715
101	3475784928
102	7457483929
102	8734648483

Handling of Multi-value Attribute



■ Solution: Two Tables

Primary Key

Foreign Key

Cno	C_Name	City
100	Rahat Bhatia	Patiala
101	Ruhani Sharma	Jalandhar
102	Rishan	New Delhi

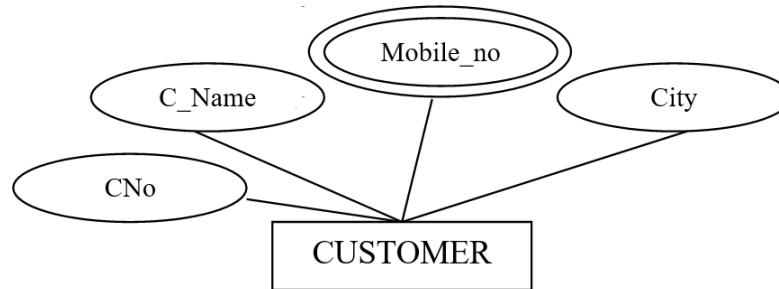
Table: Customer

Primary Key	
Cno	Mobile_No
100	1876115046
100	8739593711
100	8739593715
101	3475784928
102	7457483929
102	8734648483

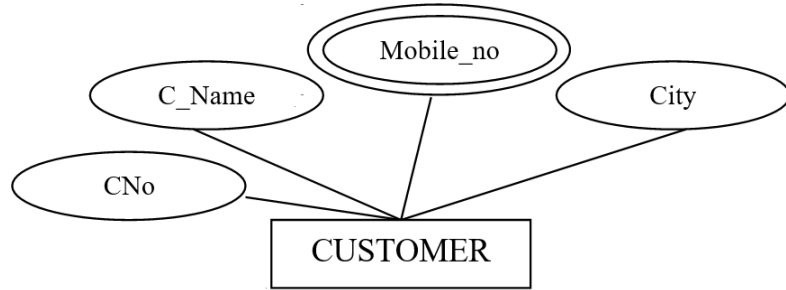
Table: Cust_Mob

Solution for Handling of Multi-value Attribute

- A multivalued attribute M of an entity E is represented by a separate table EM.
- A table EM has attributes corresponding to the primary key of E and an attribute corresponding to multivalued attribute M.



Handling of Multi-value Attribute

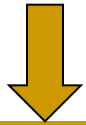
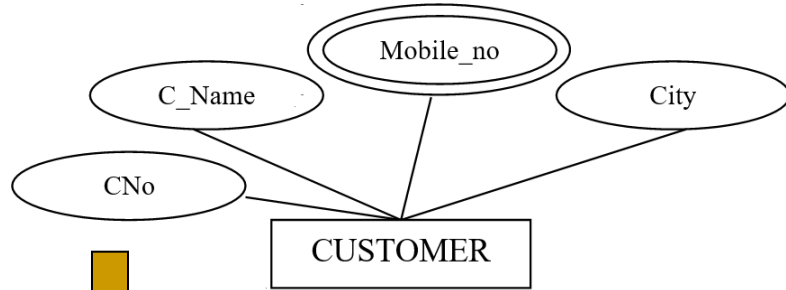


- Solution
- Two Tables

Cno	C_Name	City
100	Rahat Bhatia	Patiala
101	Ruhani Sharma	Jalandhar
102	Rishan	New Delhi

Cno	Mobile_No
100	1876115046
100	8739593711
100	8739593715
101	3475784928
102	7457483929
102	8734648483

Handling of Multi-value Attribute



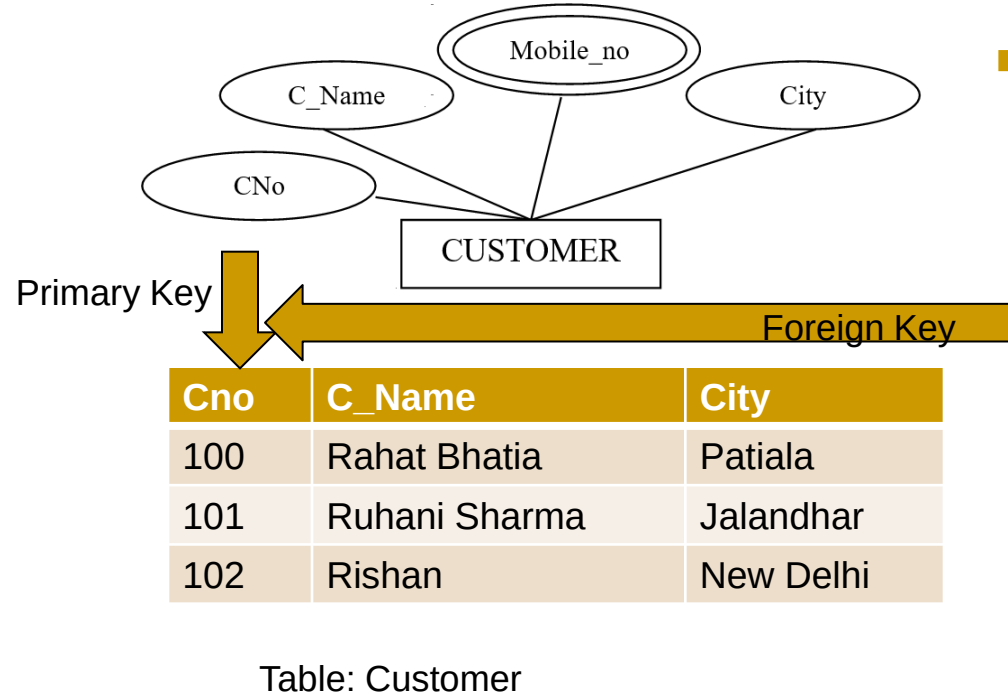
Primary Key

Cno	C_Name	City
100	Rahat Bhatia	Patiala
101	Ruhani Sharma	Jalandhar
102	Rishan	New Delhi

- Solution
- Two Tables
Primary Key

Cno	Mobile_No
100	1876115046
100	8739593711
100	8739593715
101	3475784928
102	7457483929
102	8734648483

Handling of Multi-value Attribute

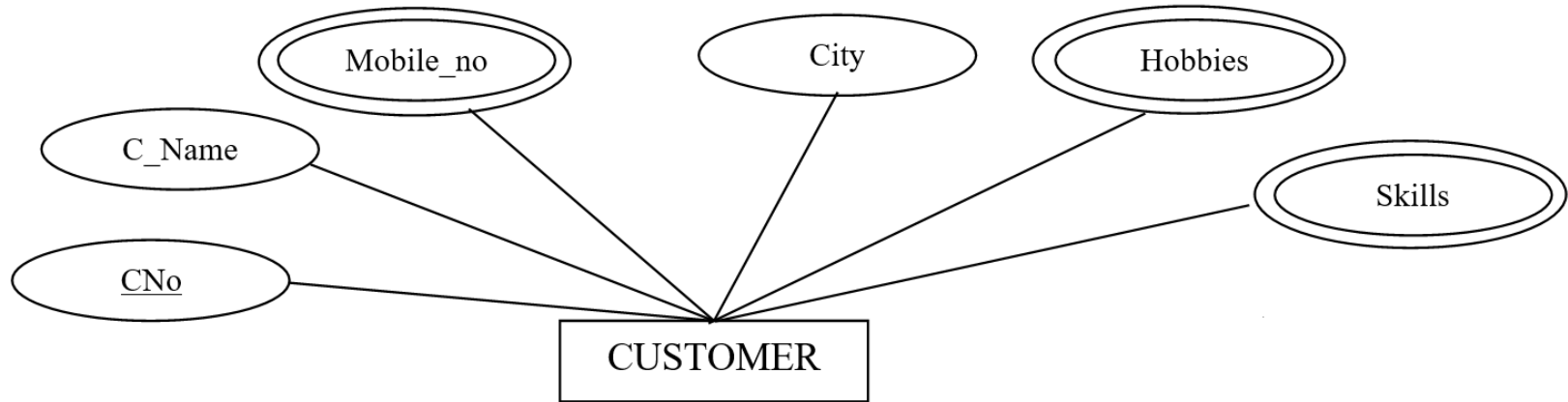


■ Solution: Two Tables

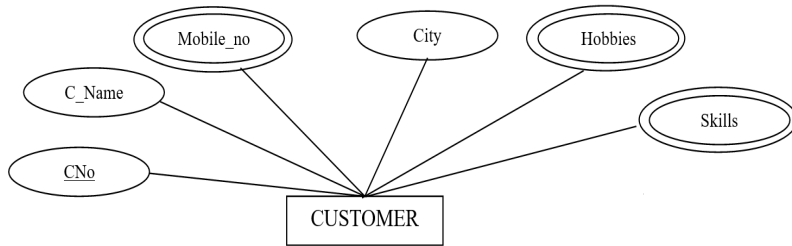
Primary Key	
Cno	Mobile_No
100	1876115046
100	8739593711
100	8739593715
101	3475784928
102	7457483929
102	8734648483

Table: Cust_Mobile

Exercise



Solution

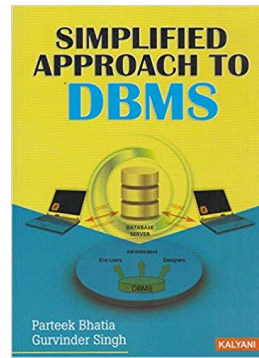


Tables: 04

- Customer
 - ❑ Cno, Cname, City
- Cust_Mobile
 - ❑ Cno, Mobile_No
- Cust_Hobby
 - ❑ Cno, Hobby
- Cust_Skill
 - ❑ Cno, Skill

Conversion of E-R Diagram to Tables

Handling of Derived Attributes

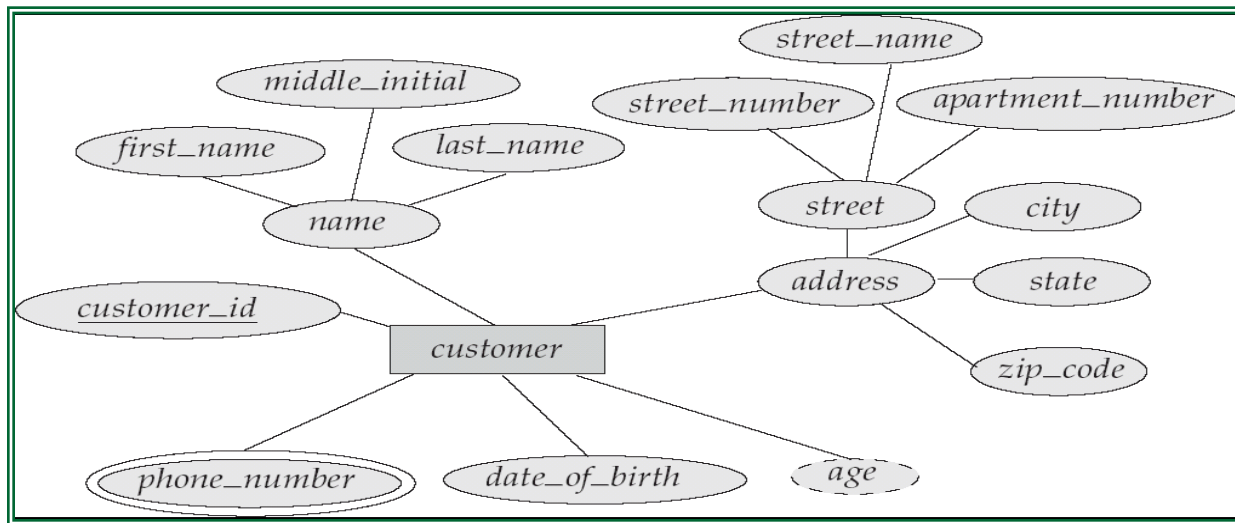


Derived Attribute

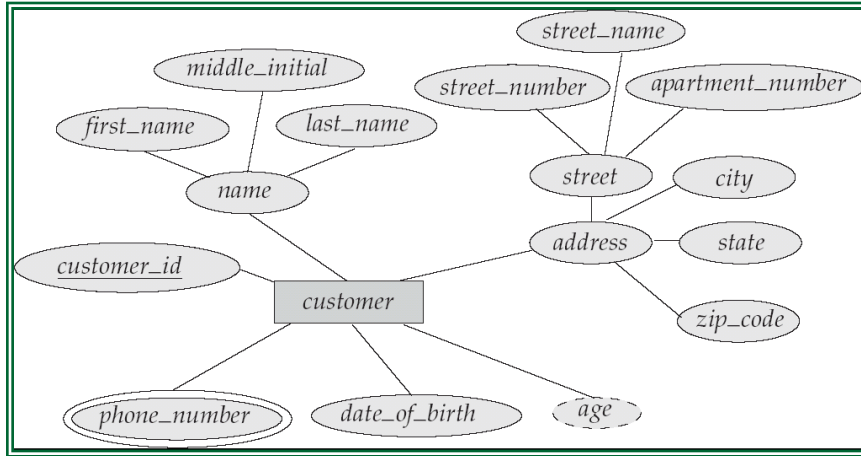
- ❑ Attribute that can be calculate based on other attributes.
- ❑ Example: Age can be calculated based on Date of Birth



Example

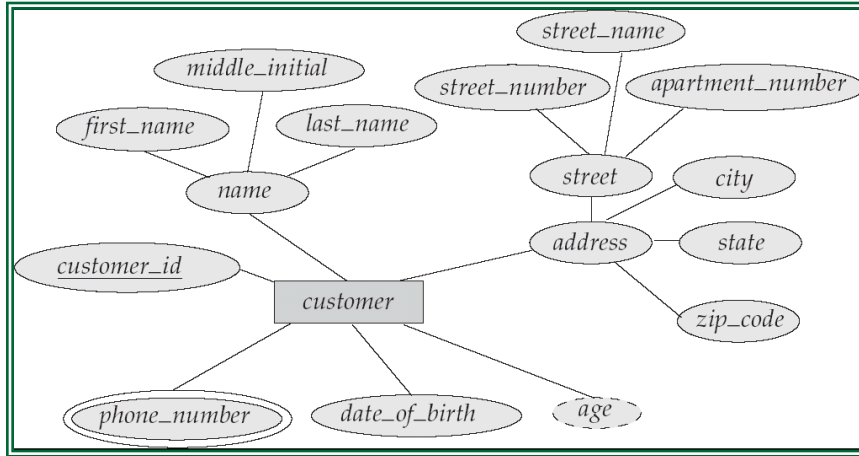


Example



- No need to create a column for age.
- Its value will be derived from date_of_birth column
- The query for this is:
SELECT
(SYSDATE- DATE_OF_BIRTH)/365 AGE
FROM CUSTOMER;

Conversion of Entity Set to Tables



- Table: CUSTOMER
- Columns:
customer_id, first_name,
Middle_initial, last_name, stree_number,
street_name, apartment_number, city,
state, zip_code, date_of_birth
There will be 11 columns
- Table: Customer_phone
- Columns:
Customer_id, phone_number